

Warehouse AI Analysis

Analyze today's warehouse backlog

To: CEO
From: Senior Retail Warehouse Operations Consultant
Subject: Operational Risk Assessment – Pending Parcel Backlog Analysis
Date: [Current Date]

Executive Summary

The warehouse network is holding **₹17.8 Crore** in pending inventory value across **4,031 parcels**. While the network average age is manageable, two warehouses—**WH-VJA-SSKL-ENIKEPADU** and **WH-HYD-SSKL-KSMTPR**—are in a state of operational stagnation. **57.6%** of VJA's pending parcels are over 10 days old, and **HYD's** average age of 11.9 days indicates a systemic, chronic delay rather than isolated incidents.

The primary bottleneck is the **SAREES-LF (Ladies)** section, which accounts for **67.9%** of its own parcels as pending and holds **₹5.98 Crore** in unresolved value. The sourcing hub of **Surat** is the epicenter of the backlog, contributing approximately **70%** of all pending records.

A single **44-day-old parcel** at **WH-BNGR-SSKL-KANKPR** with **zero invoice value** represents a critical data integrity failure. Immediate escalation is required to prevent inventory write-off and vendor disputes.

Critical Exceptions

1. Oldest CNS (Critical Non-Stock / Priority Parcel)

- Observation:** A single consignment (CNS No. 773353) from **Prasad Novelties** (Bangalore) has been pending for **44 days** at **WH-BNGR-SSKL-KANKPR**.
- Evidence:** Warehouse age = 44 days. Vendor invoice number = "NA". Vendor invoice amount = **₹0.00**. Purchaser = naga.srinivas.gunda.
- Business Impact:** This parcel is a **ghost record**. It cannot be processed for payment, put-away, or dispatch. It artificially inflates the pending count and blocks warehouse capacity. If lost, it represents a **100% write-off**.
- Priority:** **CRITICAL – Immediate Escalation Required.**
- Recommended Investigation:** Physically locate this parcel. Determine if it is a sample, a return, or a data entry error. The purchasing team must resolve the missing documentation within **24 hours**.

2. Highest Value Pending Parcel

- Observation:** The highest single-value consignment is from **SHRI SHARDA SAREES** (Surat) at **WH-HYD-SSKL-KSMTPR**, valued at **₹1,95,520** across **10 parcels**, aged **31 days**.
- Evidence:** CNS No. 776891. Warehouse age = 31 days. Parcel count = 10.
- Business Impact:** Nearly **₹2 Lakhs** of inventory is not on the sales floor. At 31 days,

this stock is depreciating (fashion goods) and occupying premium space. This represents a direct cash flow blockage.

- ****Priority:** **HIGH - Immediate Processing Required.****
- ****Recommended Investigation:**** Verify if the delay is due to a quality check (QC) bottleneck or a missing PO approval for this specific bulk shipment.

3. Highest Risk Warehouse

- ****Observation:** **WH-VJA-SSKL-ENIKEPADU**** is the highest risk location.
- ****Evidence:****
 - Highest absolute pending records: ****1,057****.
 - Highest pending value: ****₹6.96 Crore****.
 - Highest count of parcels over 20 days: ****210****.
 - ****57.6%**** of its pending parcels are over 10 days old.
 - ****10** out of **13**** outlier CNS records belong to this warehouse.
- ****Business Impact:**** This single warehouse holds ****39%** of the total network pending value**. The process for handling these parcels has likely broken down. This is a ****cash flow and customer churn crisis****.
- ****Priority:** **CRITICAL - Systemic Failure.****
- ****Recommended Investigation:**** Investigate **why** 210 parcels are over 20 days old. Is it a sorting machine failure? A staffing shortage? A specific carrier not picking up? Do not assume it is just "high volume."

4. Highest Risk Vendor

- ****Observation:** **Jiravala Textiles**** (Surat) is the highest risk vendor.
- ****Evidence:****
 - Highest pending records: ****167**** (100% of their parcels are pending).
 - Average age: ****19.3 days****.
 - Max age: ****24 days****.
 - Pending value: ****₹39.66 Lakhs****.
- ****Business Impact:**** This vendor is likely to file a claim for delayed acceptance or damage. The 100% pending rate suggests a ****systemic rejection or hold**** on their entire shipment batch.
- ****Priority:** **HIGH - Vendor Relationship Risk.****
- ****Recommended Investigation:**** Physically inspect the receiving area for Jiravala Textiles. Is there a pile of quarantined goods? Is there a quality dispute or a missing document holding up the entire batch?

5. Highest Risk Section

- ****Observation:** **SAREES-LF (Ladies)**** is the highest risk section.
- ****Evidence:****
 - ****1,382**** pending records out of ****2,034**** total parcels (67.9% pending rate).
 - Pending value: ****₹5.98 Crore****.
 - Max age: ****37 days****.
- ****Business Impact:**** This section represents the largest single queue of unresolved work. It is consuming the majority of warehouse investigation labor. The 37-day max age implies at least one customer has been waiting over a month for resolution.
- ****Priority:** **CRITICAL - Process Redesign Required.****
- ****Recommended Investigation:**** Audit the last 50 resolved and 50 unresolved SAREES-LF parcels. Determine if the bottleneck is at **receiving** (wrong SKU), **quality check** (damage), or **system update** (barcode scan failure).

Operational Bottlenecks

1. Surat Sourcing Hub Congestion

- **Observation:** Surat-based vendors account for approximately **70%** of all pending records (~900 records) and **₹2.85 Crore** in pending value.
- **Evidence:** Vendor summary shows 10 of the top 12 backlog vendors are from Surat (Jiravala, Sri Shyam, Chitrakoot, VINAYAKA, LF NEW VENDOR, Sanskriti, LITCHI, VIRA FASHION, etc.).
- **Business Impact:** The receiving team in Surat is either understaffed, lacks space, or has a specific process failure (e.g., a broken scanner, a rejected quality batch) that is cascading across multiple vendors.
- **Priority:** **HIGH.**
- **Recommended Investigation:** Physically inspect the Surat receiving area. Is there a single piece of equipment down? Is there a pile of "quarantined" goods from Jiravala or Sri Shyam?

2. WH-HYD Systemic Delay

- **Observation:** WH-HYD-SSKL-KSMTPR has the **highest average age (11.9 days)** in the network.
- **Evidence:** 412 parcels over 10 days, 154 over 20 days. Average age is past the 10-day threshold.
- **Business Impact:** Unlike VJA (which has a mix of old and new), HYD's entire pipeline appears slow. This is a chronic, not acute, issue.
- **Priority:** **HIGH.**
- **Recommended Investigation:** Compare the workflow at HYD to the "Little Kanchi" warehouse, which has a similar profile but much faster throughput (avg age 3.4 days). Identify the process step causing the delay.

Root Cause Analysis

1. Missing LR (Lorry Receipt) Concentration

- **Observation:** The `not\_mapped` field is **0** for every single vendor in the dataset.
- **Conclusion:** Based on supplied data, there is **no backlog** caused by missing LR mapping. The bottleneck is not in the initial receipt registration; it is in the subsequent processing (inspection, quality check, system put-away).

2. Box Mapping Concentration

- **Observation:** The dataset does not contain a field for "box mapping" status or errors.
- **Conclusion:** We cannot identify any box mapping trends from this data.
- **Recommended Investigation:** The WMS must be queried for a separate report on "Box Mapping Accuracy" or "Unmapped Boxes" to identify if this is a hidden bottleneck.

3. Data Quality Issues (Root Cause of Ghost Inventory)

- **Observation:** Multiple records have critical data fields marked as "NA" or ₹0.00.
- **Evidence:**
 - **Prasad Novelties** (44 days): Invoice = NA, Amount = ₹0.00.
 - **QUEENS ART JEWELLERY** (34 days): Invoice = NA, Amount = ₹0.00.
 - **VIRA FASHION** (Multiple entries): Purchaser = NA, PO Numbers = Blank.
 - **Praveen Silk Sarees** & **VENUS CREATION**: Avg_age = 0.0, Max_age = 0.0 (data entry timestamp errors).
- **Business Impact:** These "ghost" records cannot be processed for payment or put-away. They artificially inflate the pending count and block warehouse capacity.
- **Priority:** **CRITICAL – Data Hygiene Failure.**
- **Recommended Investigation:** The Operations Manager must escalate these cases to the Purchasing and Finance heads immediately. The "NA" data fields must be corrected within 24 hours

to allow processing.

Financial Exposure

- **Total Network Pending Value:** ₹17.8 Crore.
- **Critical Bucket (Above 20 days):** 366 parcels (9.1% of total parcels). The financial value tied to these parcels is likely disproportionate, as many high-value items (e.g., ₹1.95L, ₹1.53L) are in this bucket.
- **VJA Concentration:** ₹6.96 Crore (39% of total) is tied up in a single warehouse.
- **SAREES-LF Concentration:** ₹5.98 Crore (33.6% of total) is tied up in a single section.
- **Risk:** This capital is not turning into revenue. If these parcels are lost or damaged, the financial hit is substantial.

Data Quality Issues

1. **Division/Section Mismatch:** Records exist with `"division": "NA"` paired with `"section": "LADIES"`. This indicates a breakdown in master data at the point of receipt.
2. **Zero-Value Pending Records:** 5 records (NA/LADIES and NA/SHOP IN SHOP) have a pending value of ₹0.00. These are system ghosts that need to be purged.
3. **Zero-Age Anomaly:** **Praveen Silk Sarees** and **VENUS CREATION** have an `avg_age` and `max_age` of 0.0. This indicates data entry errors or records created without a timestamp.

Today's Priority Investigation

1. **Locate CNS 773353 (Prasad Novelties, 44 days):** Physically find this parcel. Resolve the missing invoice and value.
2. **Unblock SHRI SHARDA SAREES (31 days, ₹1.95L):** Process this 10-parcel consignment immediately.
3. **Audit VJA Receiving Process:** Investigate why 210 parcels are over 20 days old. Is it a specific vendor hold or a process failure?
4. **Correct "NA" Data Fields:** Escalate to Purchasing and Finance to resolve missing invoice numbers and PO numbers for all outlier records.

Recommended Actions

Short Term (Next 48 Hours)

- **Physically locate** the 44-day-old parcel at BNGR.
- **Process** the SHRI SHARDA SAREES consignment (₹1.95L) at HYD.
- **Correct** all "NA" data fields for the top 10 outlier records.
- **Escalate** Jiravala Textiles (167 pending records) to the vendor management team.

Medium Term (Next 2 Weeks)

- **Root cause analysis** at VJA: Determine if the bottleneck is equipment, staffing, or process.
- **Process flow audit** at HYD: Compare to the "Little Kanchi" benchmark.
- **Surat dock audit:** Inspect the receiving area for quarantined goods from top backlog vendors.
- **Purge** zero-value ghost records from the system.

Long Term (Next Quarter)

- **Implement a triage process** for high-value, low-volume sections (PATTU, PATTU-DHOTIS).
- **Enforce data entry standards** at the point of receipt to eliminate "NA" fields.
- **Replicate the "Little Kanchi" process** across all warehouses.

Additional Data Required

1. **Box Mapping Report:** A separate report from the WMS on "Box Mapping Accuracy" or "Unmapped Boxes."
2. **Value-Weighted Aging Analysis:** The exact monetary value of the "above 20 days" bucket, not just parcel counts.
3. **Staffing vs. Throughput Data:** Man-hours per warehouse vs. parcels processed per day (to identify if the bottleneck is manpower).
4. **Quality Hold Report:** A list of parcels currently on "quality hold" and the reason for the hold.

Confidence

High. The analysis is based entirely on the supplied dataset. All conclusions are data-driven. The recommended investigations are designed to validate the root causes before any assumptions are made.